



COURSE SYLLABUS						
Cours e Code	INTE 402	Course Title	Capstone Project 2		Course Credit	3
Course Description		This course is the continuation of Capstone Project 1. It focuses on the implementation, testing, documentation, and final presentation of the approved IT project proposal. Students are expected to apply project management methodologies, software engineering principles, and technical skills to complete a functional system or application that addresses real-world problems. The course emphasizes teamwork, problem-solving, and adherence to professional and ethical standards in project development. At the end of the course, students will present their completed projects for evaluation and defense.				
Pre-Requisites		INTE 303	Co- Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



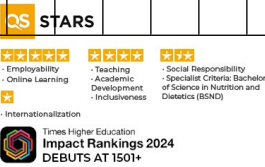
INSTITUTIONAL LEARNING
OUTCOMES
(ILO)

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
		Apply knowledge of computing, science, and mathematics appropriate to the discipline.								
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution									





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TAGUIG CAMPUS
Bachelor of Science in Information



	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.									
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.									

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
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Website: www.pup.edu.ph | Inquiries: https://bit.ly/PUPSINTA

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QS WORLD UNIVERSITY RANKINGS

2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

WURI
The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI APPLICATION

Rank 39
STUDENT SUPPORT ENGAGEMENT

Rank 50
SYMBOL/ PROMOTION

QS STARS

★★★★★
Employability
★★★★★
Online Learning
★
Internationalization

★★★★★
Teaching
★★★★★
Academic Development
★★★★★
Inclusiveness

★★★★★
Social Responsibility
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+



	Integrate IT-based solutions into the user environment effectively.								
	Assist in the creation of an effective IT project plan.								
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession								
	Understand best practices and standards and their applications.								
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.								
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.								
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.								
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.								
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development								

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	CLO1: Apply appropriate software development methodologies in the implementation of the approved IT project proposal.								
	CLO2: Develop a functional and reliable IT-based system that addresses a real-world problem or organizational need.								





	CLO3: Utilize project management tools and techniques to effectively plan, monitor, and control project milestones and deliverables.								
	CLO4: Demonstrate proficiency in using programming languages, frameworks, and databases relevant to the system being developed.								
	CLO5: Conduct systematic testing, debugging, and evaluation to ensure software quality, security, and performance.								
	CLO6: Prepare comprehensive project documentation following academic and industry standards.								
	CLO7: Communicate project results effectively through oral presentation and system demonstration before a technical panel.								
	CLO8: Exhibit professional ethics, teamwork, and responsibility throughout the development, implementation, and defense of the capstone project.								





OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	- Recall and refine approved Capstone 1 proposal - Develop detailed work plan and set milestones	- Review of approved proposal - Project scheduling tools (Gantt chart, Trello, Jira)	- Orientation and consultation meeting - Workshop on project timeline setup	- Online discussion of milestones and expectations - Live tutorial on planning tools	- Upload refined proposal and updated project plan - Independent milestone planning	- Validated project requirements - Project timeline				
3	Begin system development setup	Software tools setup, environment	Coding lab session	Live code review	 Upload 2025 7th in the Philippines 92nd in Southeast Asia 541-560 in Asia	 Configure				



		configuration			documentation	development environment				
4	Implement core backend functionalities	Backend architecture, API setup, logic flow	Coding and mentoring session	Real-time debugging support	Backend coding task	Partial backend module				
5	Apply version control and collaborative coding	Git/GitHub workflow, branching strategy	Git collaboration activity	Online demo on Git usage	Self-paced commits & merges	Git repository submission				
6	Develop and test core system modules	CRUD operations, database connectivity	Hands-on lab	Q&A troubleshooting	Submit weekly code progress	Tested backend modules				
7	Design and implement user interface	UI/UX principles, responsive design	Design workshop	Online design critique	Upload mockups and prototype	UI design evaluation				



8	- Integrate frontend and backend modules - Conduct functional and unit testing	- RESTful APIs, data validation, integration testing - Testing methods, test case design	- Integration lab session - Testing activity	- Integration guidance via meet - Q&A on testing tools	- Upload integrated build - Independent bug tracking	- Proto type dem o - Unit test resul ts				
9	MIDTERM WRITTEN EXAMINATION									
10	Conduct system debugging and optimization	Code review, performance tuning	Debugging session	Live support for bug fixing	Continuous integration updates	Debuggin g report				
11	Perform database and security validation	Data integrity, input validation, encryption basics	Database test session	Security best practices lecture	Upload SQL scripts	Database validation report				
12	Prepare technical documentation (Ch. 1-5 draft)	Documentation standards (IEEE, APA), formatting	Writing workshop	Online formatting Q&A	Self-paced document writing	Draft document ation (Ch. 1-5)				
13	Conduct User Acceptance Testing (UAT)	UAT process, user feedback forms	User testing with respondents	Online feedback consolidation	Upload UAT report	UAT summary report				
14	Finalize system and documentation	Revisions and polishing	Consultation and system fine-tuning	Q&A on final edits	Upload final version	Final system ready for defense				
15	Prepare for project defense	Presentation skills, defense format	Mock defense and evaluation	Online rehearsal and feedback	Prepare presentation slides	Mock defense performan ce				
16	Conduct mock defense	Defense simulation, Q&A handling	Practice defense session	Online feedback and critique	Revise based on mock defense	Mock defense				



						performanc e				
17	Apply revisions based on panel feedback	System and document revisions	Consultation and correction check	Online monitoring	Upload revised outputs	Revised document ation and system				
18	FINAL WRITTEN EXAMINATION (Final Presentation, Practical Exam)									



		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		• Quizzes					
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FINAL EXAMINATION

LIST OF REFERENCES

BOOKS:

WEBSITE:

1.

Tenhunen, S., Männistö, T., Luukkainen, M., & Ihtola, P. (2023). *A systematic literature review of capstone courses in software engineering* — this paper synthesizes the characteristics, durations, team sizes, assessment strategies, and project models in existing software engineering capstone courses. ([arXiv](#))

2.

Estell, J. K., & Hurtig, J. K. (2014). *Adopting Best Corporate Practices for Capstone Courses: A Case Study at Ohio Northern University* — describes use of industry-style project management, documentation standards, and formative/summative assessments in capstone design courses. ([ResearchGate](#))

3.

Majanoja, A.-M. et al. *Reflections on Teaching Software Engineering Capstone Course* — examines practical issues in organizing capstone projects in software and computing curricula, with recommendations for active learning and student responsibility. ([UTUPub](#))

4.

Matthies, C. (2018). *Scrum2Kanban: Integrating Kanban and Scrum in a University Software Engineering Capstone Course* — explores how combining Scrum and Kanban methods can enhance agility and learning in capstone teams. ([arXiv](#))

5.

“Software is the easy part of Software Engineering” – *Lessons and Experiences from A Large-Scale, Multi-Team Capstone Course* — discusses challenges of multi-team coordination, architecture, and communication in large-scale student projects. ([arXiv](#))

6.

Laplante, P. A., DeFranco, J. F., & Guimaraes, E. (n.d.). *Evolution of a Graduate Software Engineering Capstone Course—A Course Review* — charts changes over time in capstone design, agile adoption, and assessment practices. ([ijee.ie](#))

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Teaching
Academic Development
Social Responsibility
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Times Higher Education Impact Rankings 2024
DEBUTS AT 1501+



7. Wagner, J. L. (2023). *Best Practices in Syllabus Design* — while not specific to capstone courses, this article outlines key considerations for syllabus clarity, structure, and alignment of outcomes, useful for designing your weekly plan. ([PMC](#))

8. Bukhari, N. (2024). *Designing and Implementing the Undergraduate Capstone* — critiques common misalignments in capstone curriculum design and offers guidance on aligning project-based learning with core academic outcomes. ([seaairweb.info](#))

CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.





The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is “INC” grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don’t tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student’s own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a

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Academic
Development
Inclusiveness
Internationalization

Specialist Criteria: Bachelor
of Science in Nutrition and
Dietetics (BSND)

DEBUTS AT 1501+



	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
- Capstone Project 2 focuses on the development, implementation, evaluation, and defense of an IT-based solution based on the approved Capstone Project 1 proposal. Students must design and deploy a fully functional system demonstrating technical proficiency, innovation, and professionalism. - Project progress is monitored through adviser consultations and project management tools, with documentation following the PUPT Dissertation Manual and APA 7th Edition format. - The course culminates with a midterm presentation, final defense, and submission of all required deliverables.	Mid Term	Final Term
	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper
	Midterm Examinations 30% 100%	Final Examinations 30% 100%
Final Grade = (Midterm Grade + Final Term Grade) / 2		
CLASS INFORMATION		FACULTY INFORMATION
Course, Year and Section: BSIT 4-1		Name of Faculty: Almirañez, Gecilie C.
Day and Time: F 01:00pm-4:00pm		Consultation Time:
Room: TGA205		Office Tel. No./ Mobile Phone No. 2025

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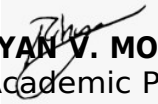
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School Year: 2025-2026 Semester: First Semester	E-Mail Address:
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Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
Gecilie C. Almirañez Faculty Member	 DR. RHYAN V. MOLINAR Head of Academic Programs	 DR. MARISSA B. FERRER Campus Director	 PROF. PASCUALITO B. GATAN Vice President for Campuses

